# Ritvik Sai Narayan

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## Education

**Massachusetts Institute of Technology (MIT)** Expected May 2031
*Ph.D in Physics, Astrophysics Division*

**University of Wisconsin–Madison** May 2026
*B.S. in Astronomy–Physics (Honors) & Economics*
*Graduated with Distinction*
Thesis: A Coupled MESA–GYRE Framework for Modeling Star-Companion Tidal Interactions

## Publications

### First Author

5. **Narayan, R. S.**, Livesey, J. R., & Becker J. *Exomoonlight Sonata: On The Fate of Exomoons Around Migrating White Dwarf Planets*, in prep.
4. **Narayan, R. S.**, & Soares-Furtado, M. *starfuSED: A Self-Contained SED Fitter for Single and Binary (Sub)Stellar Systems*, in prep.
3. **Narayan, R. S.**, Soares-Furtado, M., & Townsend, R.H.D. *The Tale of a Hungry Subgiant and Its Brown Dwarf: Interior Radiative Damping Dominates the Tidal Evolution of TOI-5882*, 2026, *ApJL*, 1005, L35. [2605.04141]
2. **Narayan, R. S.**, Soares-Furtado, M., Limbach, M. A., et al. *Twinkle Twinkle Little Star, Roman Sees Where You Are: Predicting Exoplanet Transit Yields with a 30-day Survey of the Rosette Nebula with the Nancy Grace Roman Space Telescope*, 2026, *AJ*, 171, 282. [2510.23708]
1. **Narayan, R. S.**, Linck, E., Mathieu, R.D., et al. *WIYN Open Cluster Study. XCVII. An Extended Radial-Velocity Survey and Spectroscopic Binary Orbits in NGC 188*, 2026, *AJ*, 171, 102. [2509.12315]

### Contributing Author

4. Aloisi, R., Vanderburg, A., Soares-Furtado, M., et al. (including **Narayan, R. S.**) *VOYAGERS Survey Paper 2: A Giant Planet Candidate Orbiting TIC 20897763, A Metal-Poor Gaia-Enceladus Sausage Star*, in prep.
3. Kroft, M., Beatty, T. G., Salzer, J. M., et al. (including **Narayan, R. S.**) *GJ 523b is a Massive, 150 Myr-old mega-Earth on a Likely Polar Orbit*, submitted to *AJ*. [2603.24682]
2. Distler, A., Soares-Furtado, M., Mann, A. W., et al. (including **Narayan, R. S.**) *TESS Hunt for Young and Maturing Exoplanets (THYME) XIV: A Comoving-Based Age Constraint for KELT-20*, 2026, *AJ*, 171, 248. [2603.01313]
1. Sheffler, J. K., Clark, M., Soares-Furtado, M., et al. (including **Narayan, R. S.**) *A Multi-Method Age Determination for the Ursa Major Moving Group*, 2026, *AJ*, 171, 143. [2601.08985]

## In the News

*( † Denotes media coverage of primary-authored research.)*

- **BBC Sky at Night Magazine**: *Astronomers found a 'mega Earth' that shouldn't exist. This*

*alien world is breaking the rules of planet building*. (2026)

- **The New York Times**†: *This Star Just Ate a Planet, and It's Not Done Yet*. (2026)
- **Astrobiology**†: *Predicting Exoplanet Transit Yields In The Rosette Nebula With The Nancy Grace Roman Space Telescope*. (2025)

## Software

- **GYRE** — Co-developer for v9.0. GYRE is a stellar oscillation code. Given an input stellar model, GYRE calculates the eigenfrequencies and eigenfunctions for the normal oscillation modes of the model.
- **starfuSED** — Lead Developer. starfuSED is a Python package for SED (Spectral Energy Distribution) fitting of single and binary stellar systems, providing a complete workflow from photometry retrieval to model fitting and visualization, all with in-memory processing.
- **tidalcircularization** — Lead developer. A Python interface for fitting the tidal circularization model from *Mathieu & Meibom (2005)* to stellar binary data using an MCMC algorithm.

## Observing Proposals

- **WIYN Hydra Multi-Object Spectrograph** (2024A): Co-Investigator, *A Study of Binary Populations and Alternative Pathways of Stellar Evolution in Open Clusters*, 2 nights.
- **WIYN Hydra Multi-Object Spectrograph** (2023B): Co-Investigator, *A Study of Binary Populations and Alternative Pathways of Stellar Evolution in Open Clusters*, 5 nights.

## Presentations

- *The Tale of a Hungry Subgiant and Its Brown Dwarf Companion*: Theory Group Meeting, Center for Interdisciplinary Exploration and Research in Astrophysics (CIERA), Northwestern University (invited). (2026)
- *Twinkle Twinkle Little Star, Roman Sees Where You Are: Predicting Exoplanet Transit Yields in the Rosette Nebula*: 247th Meeting of the American Astronomical Society. (2026)
- *Twinkle Twinkle Little Star, Roman Sees Where You Are: Predicting Exoplanet Transit Yields in the Rosette Nebula*: Great Lakes Exoplanets Area Meeting (GLEAM). (2025)
- *Applications of PECCARY with Asteroseismology*: Steward Observatory Galaxy Lab, University of Arizona (invited). (2025)
- *Twinkle Twinkle Little Star, Roman Sees Where You Are*: 35th Annual Wisconsin Space Conference. (2025)
- *All the Mass We Cannot See: Verifying the Presence of Dark Matter in the Milky Way*: University of Wisconsin-Madison Undergraduate Symposium. (2025)
- *Probing the Long-Period Blue Straggler Progenitor Population of NGC 188: A Radial Velocity Study using The Joker*: 244th Meeting of the American Astronomical Society. (2024)

## Awards (Total: $132,750)

- **Thomas Frank & Henry W. Kendall Fellowship** ($120,000)
  Selected from the incoming doctoral cohort for an award that provides the freedom for independent research during the first year of the PhD program. (Fall 2026)

- **Doherty Award for Excellence in Astronomy** ($750)
  This award recognizes graduating undergraduate students for exceptional academic achievement and research accomplishments throughout their undergraduate careers.
- **Peter Livingston Scholar** ($3000)
  Awarded to support research under a mentor of choice at the Department of Astronomy, University of Wisconsin-Madison. (Fall 2025)
- **Alberta A. Taylor Scholarship** ($2000)
  A one-time scholastic award presented to a student pursuing an economics degree. (Spring 2025)
- **NASA Wisconsin Space Grant Undergraduate Research Scholarship** ($3000)
  Among the first international citizens to receive this award for research supporting a NASA Mission Directorate. (Spring 2025)
- **Honors Summer Research Apprenticeship Fellow** ($3000)
  Awarded to support an intensive summer research project at the University of Wisconsin-Madison. (May 2024)
- **General Knapp Scholarship** ($1000)
  (Fall 2023)
- **University of Wisconsin–Madison Dean's List**
  (Fall 2022, Spring 2023, Fall 2023, Spring 2024, Fall 2024, Spring 2025, Fall 2025, Spring 2026)
- **Chambliss Astronomy Achievement Award**
  Honorable mention at the 244th meeting of the American Astronomical Society for exemplary research as an undergraduate. (Nov 2024)
- **Outstanding Delegate**
  UN General Assembly Third Committee (SOCHUM), Ivy League Model United Nations India, hosted by the University of Pennsylvania. (Nov 2019)
- **Batch of 1975 Award**
  Highest marks in a 2nd Language (Kannada), St. Joseph's Boys' High School. (July 2019)

## Teaching Experience

**Course Assistant and Tutor**, UW-Madison Department of Mathematics Aug 2023 – Apr 2025

- In-Classroom Peer Mentor for MATH 96: Preparatory Algebra and MATH 112: College Algebra. (Fall 2023)
- Course Assistant for MATH 340: Linear Algebra and Elementary Matrix Operations. (Spring 2024, Fall 2024, Spring 2025)
- Tutor for all courses leading to and including MATH 234: Multivariable Calculus.

## Skills

**Software & Languages**: HTCondor, Tableau, HTML5, StataSE, Bash, LaTeX, Python, Fortran.